BITS Pilani

Pilani|Dubai|Goa|Hyderabad

Artificial and Computational Intelligence
AIMLCZG557

ACI Team

AIMLCZG557 – CS#13
Reasoning over Time

Agenda for CS #13

1. Recap of CS#12
2. Sequential Decision Problems &
3. Markov Decision Process
4. Inference in Temporal Models
5. HMM
6. Inferencing types
7. Exercises

Course Plan

M1 Introduction to AI

M2 Problem Solving Agent using Search

M3 Game Playing

M4 Knowledge Representation using Logics

M5 Probabilistic Representation and Reasoning

M6 Reasoning over time

M7 Multiagent Decision making

M8 Ethics in AI

Module 6: Reasoning over time

Reasoning Over Time

A. Time and Uncertainty

B. Inference in temporal models

C. Introduction to Hidden Markov Model

D. Applications of HMM

Sequential Decision Problems & Markov Decision Process

Markov Decision Process

Sequential Problem | Partial Observability | Belief System

- Modelling sequences of random events and transitions between states over time is known as Markov chain
- Agents in partially observable environment should keep a track of current state to the extent allowed by sensors
- E.g., Robot moving in a new maze
- Agent maintains a **belief state** representing the current possible world states

Transition Model / Probability Matrix :

- Using belief state and transition model, the agent can how the world might evolve in
- next time step. To capture the degree of belief we will use Probability Theory.
- We model the change in world using a variable for each aspect of state and at each point in time.
- Current state depends only finite number of previous states.

$$\begin{array}{cc|c} C & M & \\ \hline 0.40 & 0.20 & C \\ 0.60 & 0.80 & M \end{array}$$

Markov Decision Process

Time - Uncertainty | States - Observations

- **Static World:** Each random variable would have a single fixed value
- E.g., Diagnosing a broken car

- **Dynamic World:** The state information keeps changing with time
- E.g., treating a diabetic patient, tracking the location of robot, tracking economic activity of a nation

- **Time slices:** World is observed in time slices. Each slice has a set of random variables, some observable and some not.

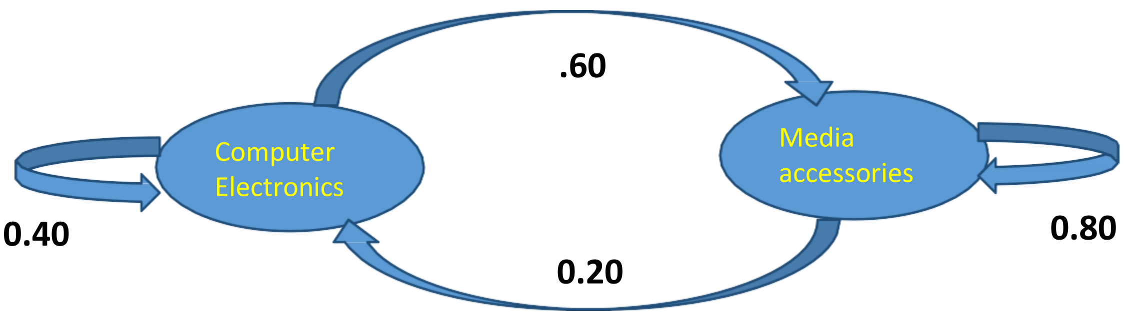
Assumption: We will assume same subset of random variables are observable in each time slice

$\mathbf{E_t}$ – set of observable random variables at time t

$\mathbf{X_t}$ – set of unobserved random variables at time t

$$\begin{array}{cc} \textbf{C} & \textbf{M} & \\ \left|\begin{array}{cc} 0.40 & 0.20 \\ 0.60 & 0.80 \end{array}\right| & & \begin{array}{c} \textbf{C} \\ \textbf{M} \end{array} \end{array}$$

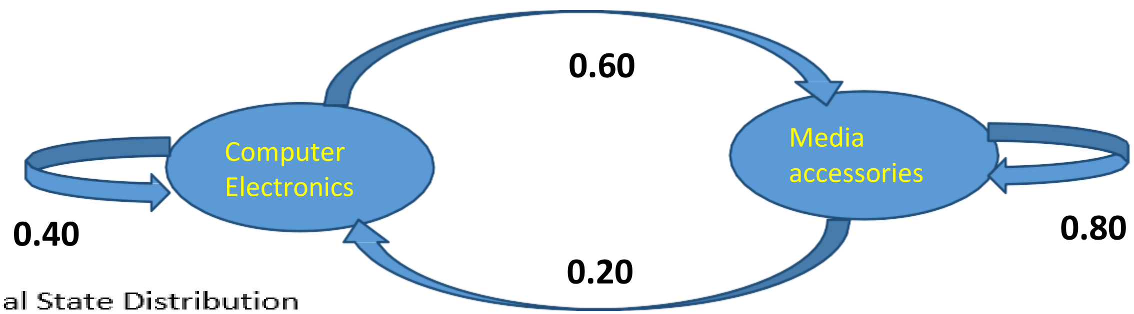
Markov Model- Example 1



Transition Model

$$\begin{array}{cc} \begin{matrix} C & \quad M \end{matrix} & \\ \begin{vmatrix} 0.40 & 0.20 \\ 0.60 & 0.80 \end{vmatrix} & \begin{matrix} C \\ M \end{matrix} \end{array}$$

Markov Model



Current State: Initial State Distribution

$$\begin{array}{c} \left| \begin{array}{c} 1 \\ 0 \end{array} \right| \end{array} \begin{array}{l} C \\ M \end{array}$$

Next State : Likely to buy Media accessories on next visit

$$\begin{array}{c} \left| \begin{array}{c} 0.40 \\ 0.60 \end{array} \right| \end{array} \begin{array}{l} C \\ M \end{array}$$

$$\begin{array}{cc} C & M \\ \end{array}$$
$$\left| \begin{array}{cc} 0.40 & 0.20 \\ 0.60 & 0.80 \end{array} \right| \begin{array}{l} C \\ M \end{array}$$

Next State : Likely to buy Media accessories on next visit

$$\begin{array}{c} \left| \begin{array}{c} 0.28 \\ 0.72 \end{array} \right| \end{array} \begin{array}{l} C \\ M \end{array}$$

Markov Process

States | Observations | Assumptions

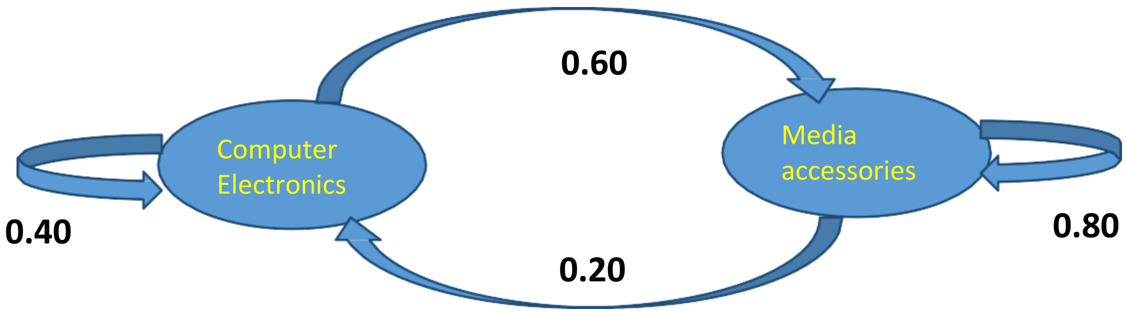
Modelling sequences of random events and transitions between states over time is known as Markov chain

Transition Model / Probability Matrix :

Current state depends only finite number of previous states. :

$$
\begin{array}{cc}
\begin{array}{cc} \mathbf{C} & \mathbf{M} \end{array} & \\
\left|\begin{array}{cc} 0.40 & 0.20 \\ 0.60 & 0.80 \end{array}\right| & \begin{array}{c} \mathbf{C} \\ \mathbf{M} \end{array}
\end{array}
$$

Markov Model



	C	M	
	0.40	0.20	C
	0.60	0.80	M

Current State: Initial State Distribution

$$\begin{array}{cc} \begin{vmatrix} 1 \\ 0 \end{vmatrix} & \begin{array}{c} C \\ M \end{array} \end{array}$$

Next State : Likely to buy Media accessories on next visit

$$\begin{array}{cc} \begin{vmatrix} 0.40 \\ 0.60 \end{vmatrix} & \begin{array}{c} C \\ M \end{array} \end{array}$$

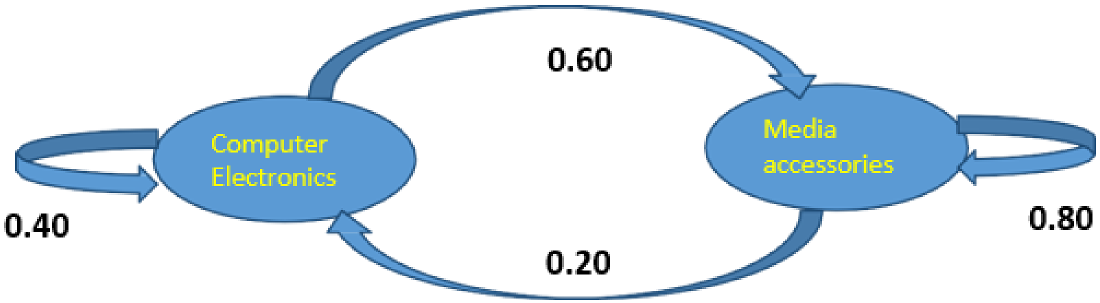
Next State : Likely to buy Media accessories on next visit

$$\begin{array}{cc} \begin{vmatrix} 0.28 \\ 0.72 \end{vmatrix} & \begin{array}{c} C \\ M \end{array} \end{array}$$

Inference in Temporal Models

Markov Model

Inference Type 1



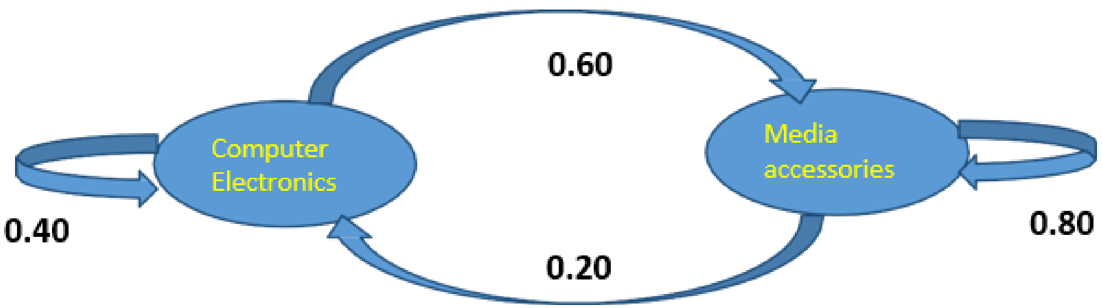
$$
\begin{array}{cc|c}
C & M & \\
\hline
0.40 & 0.20 & C \\
0.60 & 0.80 & M
\end{array}
$$

What is the probability that the purchasing behaviour of the customer is in below sequential order only?Initial Probability Matrix is P(C) = 1 , P(M) = 0
(Computer, Media, Media, Computer)

<u>Apply Bayes chain rule:</u>
P(Computer, Media, Media, Computer) = P(C) * P(M|C) * P(M|M) * P(C|M) = 0.096

Markov Model

Inference Type 2



$$\begin{array}{cc|c} C & M & \\ 0.40 & 0.20 & C \\ 0.60 & 0.80 & M \end{array}$$

What is the probability that the customer who purchased Media accessories will keep coming back to purchase media accessories in the next 2 consecutive visits only?

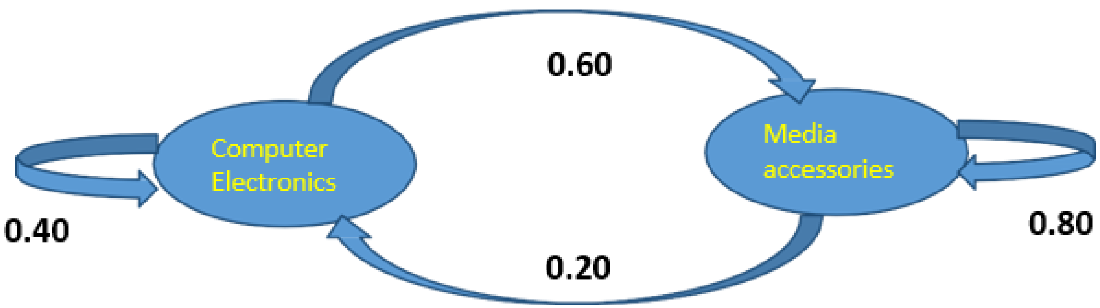
<u>Derive Initial prob values & Apply Bayes chain rule on the pattern exhibited:</u>

Initial Probability Matrix is P(M) = 1 , P(C) = 0

P(Media, Media, Media, Computer) = P(M) * P(M|M) * P(M|M) * P(C|M) = 0.128

Markov Model

Inference Type 3



$$\begin{array}{cc|c} C & M & \\ \hline 0.40 & 0.20 & C \\ 0.60 & 0.80 & M \end{array}$$

Given the evidence that the customer walked into the store and bought a computer electronics, find the expected purchase pattern in the next 3 visits

<u>Derive Initial prob values & Apply Bayes chain rule and reverse predict the combination on the most likely pattern (Similar to Viterbi Algorithm):</u>

Initial Probability Matrix is P(C) = 1 , P(M) = 0

P(Computer , X, Y, Z) = P(Computer) * P(X|Computer) * P(Y|X) * P(Z|X) = 1*0.6*0.8*0.8 → Produces max values

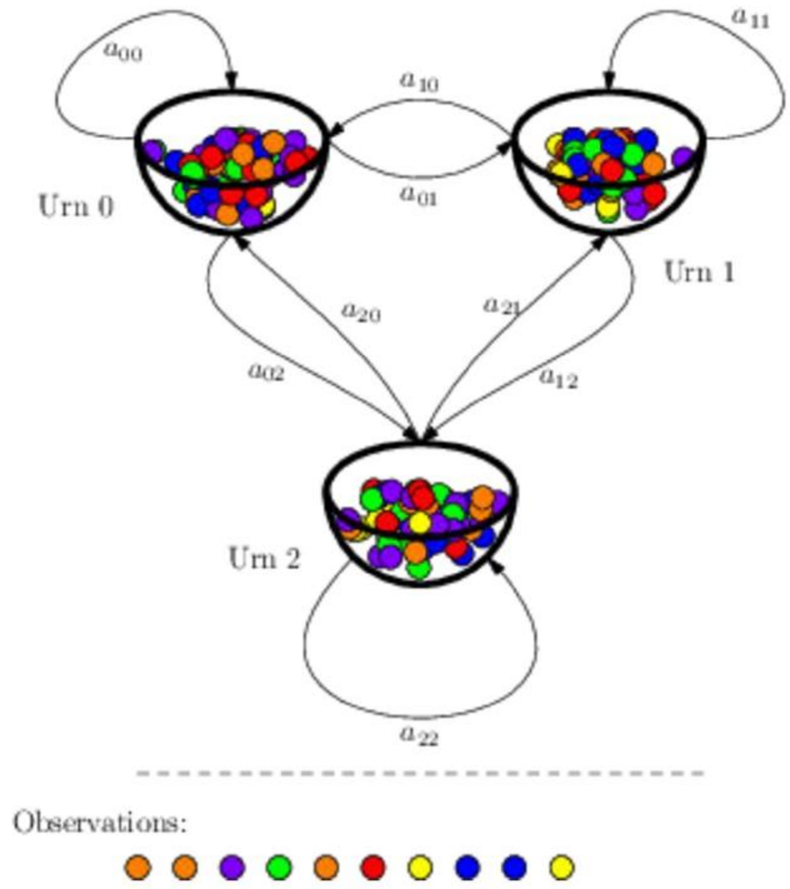
Ans : Pattern = (Computer, Media, Media, Media)

HMM

Markov Process

States | Observations | Assumptions

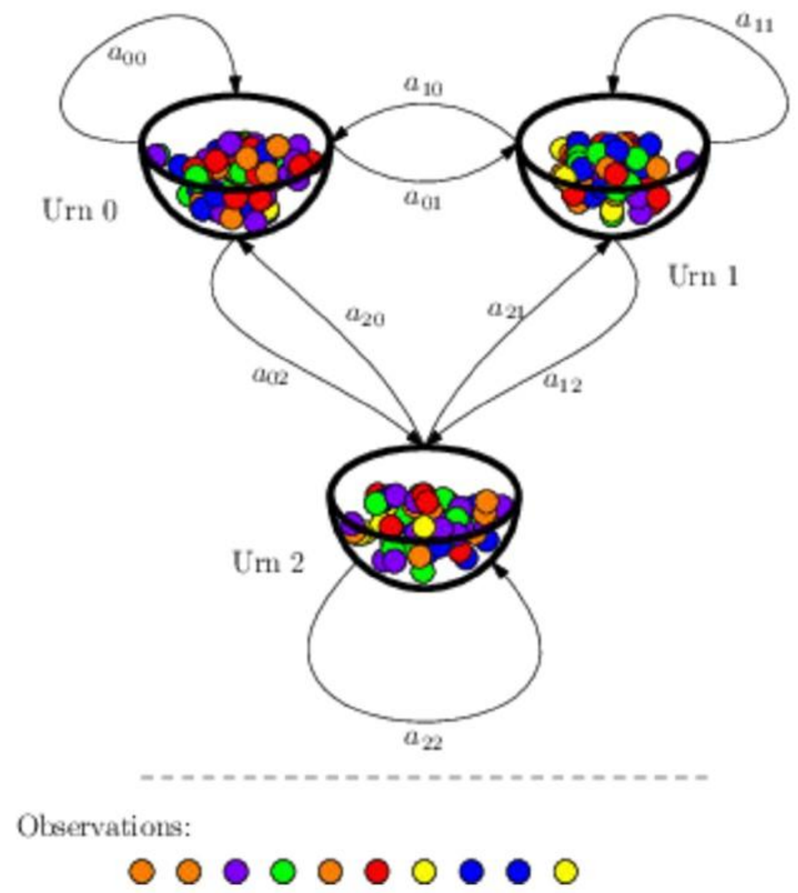
Standard Mathematical Example:
Urn & Ball Model



Markov Process

States | Observations | Assumptions

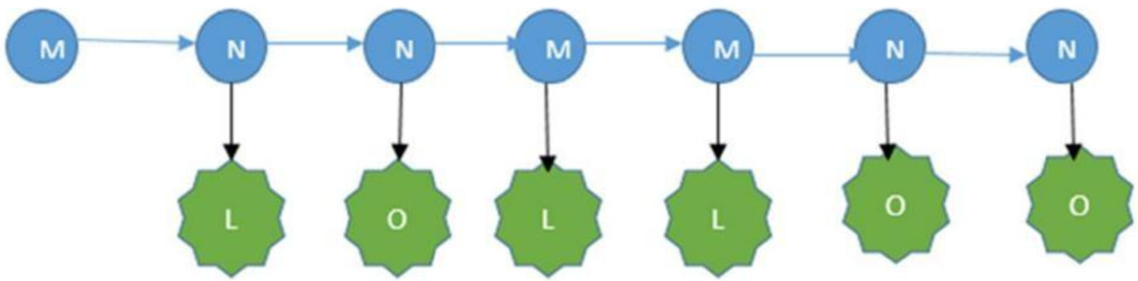
Standard Mathematical Example:
Urn & Ball Model



Hidden Markov Model

States | Observations | Assumptions

Time Slice (t)	0	1	2	3	4	5	6	$P(O_t \mid O_{t-1})$
Observed Evidence (O_t/ E_t)	-	Late	OnTime	Late	Late	Ontime	Ontime	……
Unobserved State (U_t /X_t/ Q_t)	Meeting	No Meeting	No Meeting	Meeting	Meeting	No Meeting	No Meeting	……



Transition Model / Probability Matrix

$P(U_{t-1}$ = No Meeting)	$P(U_{t-1}$ = Meeting)	←Previous
0.5	0.67	$P(U_t$ = No Meeting)
0.5	0.33	$P(U_t$ = Meeting)

Evidence / Sensor Model/ Emission Probability Matrix

$P(U_t$ = No Meeting)	$P(U_t$ = Meeting)	←Unobserved Evidence v
0.9	0.3	$P(O_t$ = OnTime)
0.1	0.7	$P(O_t$ = Late)

Hidden Markov Process

States | Observations | Assumptions

Modelling sequences of random events and transitions between states over time is known as Morkov chain

Hidden Markov Process models events as the state sequences that are not directly observable but only be approximated from the sequence of observations produced by the system

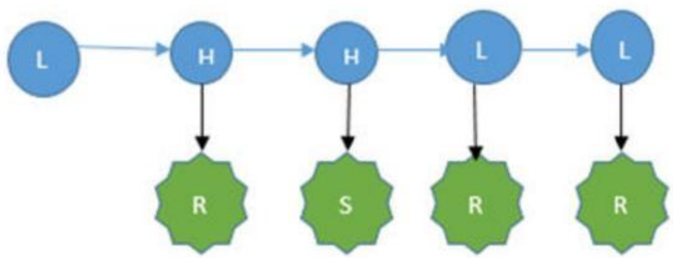
Transition Model / Probability Matrix :
Current state depends only finite number of previous states. :

Evidence / Sensor Model/ Emission Probability Matrix :
Current Evidence or Observation depends Current State of the world. Given the Current State Knowledge of the world, observation doesn't depend on history:

Hidden Morkov Model

States | Observations | Assumptions

Time Slice (t)	0	1	2	3	4	....	$P(O_t \mid O_{t-1}, O_{t-2})$
Observed Evidence (O_t)	-	Rainy	Sunny	Rainy	Rainy		
Unobserved State(U_t)	Low Pressure	High Pressure	High Pressure	Low Pressure	Low Pressure		



Transition Model / Probability Matrix

$P(U_{t-2} = LP, U_{t-1} = HP)$	$P(U_{t-2} = HP, U_{t-1} = HP)$	$P(U_{t-2} = HP, U_{t-1} = LP)$	$P(U_{t-2} = LP, U_{t-1} = LP)$	←Previous
0.2	0.40	0.85	0.5	$P(U_t = LP)$
0.8	0.60	0.15	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

Hidden Morkov Model

Filtering : P(SecondUrnIsSelected $_3$ | Red-Blue-Blue-Yellow)

$P(X_t \mid E_{1\ldots t})$

Prediction: P(FirstUrnWillbeSelected $_3$ | Red-Yellow)

$P(X_{t+k} \mid E_{1\ldots t})$

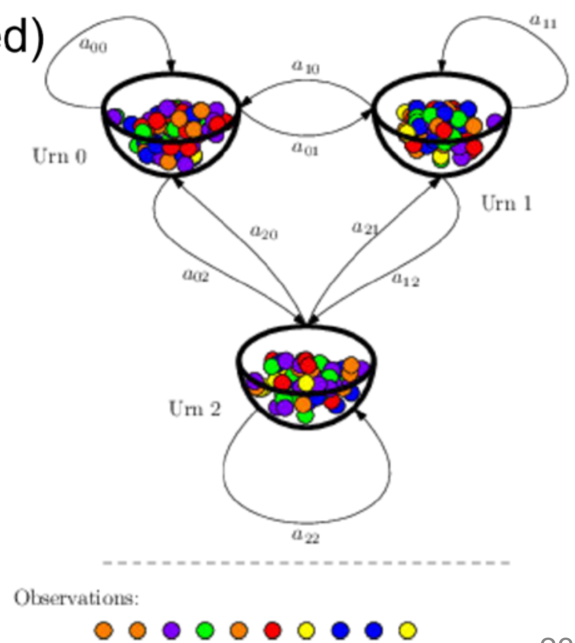
Smoothing: P(ThirdUrnWasSelected $_2$ | Red-Yellow-Red-Red)

$P(X_{k,\ o>k>t} \mid E_{1\ldots t})$

Most Likely Explanation (or) Viterbi Algorithm

P(Urn1-Urn2-Urn1 | Red-Yellow-Yellow)

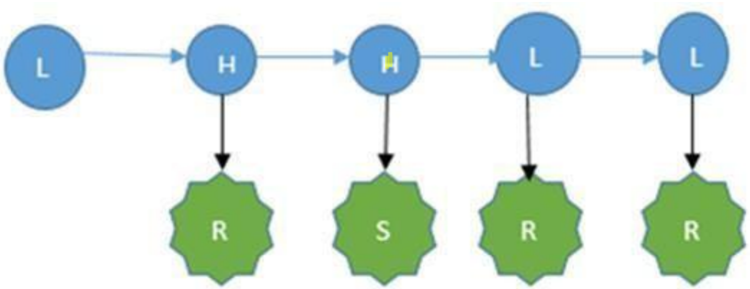
argmax $X_{1\ldots t}$: $P(X_{1\ldots t} \mid E_{1\ldots t})$



Hidden Markov Model

Filtering	Prediction	Smoothing	Most Likely Explanation
$P(X_t \mid E_{1 \dots t})$	$P(X_{t+k} \mid E_{1 \dots t})$	$P(X_{k,\ o>k>t} \mid E_{1 \dots t})$	$\text{argmax } X_{1 \dots t} : P(X_{1 \dots t} \mid E_{1 \dots t})$

In your Text book another example for these inferences is explained "Task of predicting the weather condition by a security personnel sitting in an underground secret installation by observing the state of an employee who either umbrella or don't" Kindly check it and work it out as additional practice

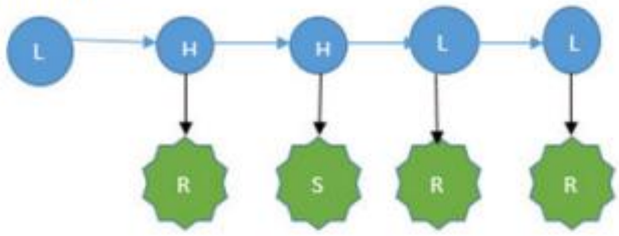


Hidden Markov Model

Inference: Type -1

Sequence Evaluation : Likely hood Computation : Forward Algorithm

Find the probability of occurrence of
this weather sequence observation: **S-S-R**



Intuition: $P(E_{1\ldots t}) = \sum_{i=1}^{N} P(E_{1\ldots t} \mid X_{1\ldots t}) * P(X_{1\ldots t}) =$

$= \sum_{i=1}^{N} \prod_{j=1}^{t} P(Ej \mid Xj) * P(Xj \mid Xj_{-1})$

Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

P(SSR)

$$= \sum_X P(SSR, X) = \sum_X P(SSR, X_1X_2X_3)$$

$$= \sum_X P(R, X_3, S, X_2, S, X_1) = \sum_X P(R \mid X_3) * P(X_3 \mid X_2) * P(S \mid X_2) * P(X_2 \mid X_1) * P(S \mid X_1) * P(X_1 \mid X_0)$$

$$= \sum_X P(R \mid X_3) * P(S \mid X_2) * P(S \mid X_1) * P(X_3 \mid X_2) * P(X_2 \mid X_1) * P(X_1 \mid X_0)$$

$$= \sum_X \prod_{j=1}^{t} P(Ej \mid X_j) * P(Xj \mid X_{j-1})$$

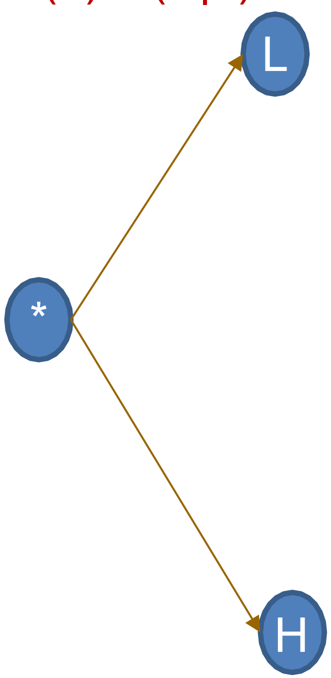
Hidden Markov Model

Forward Propagation Algorithm

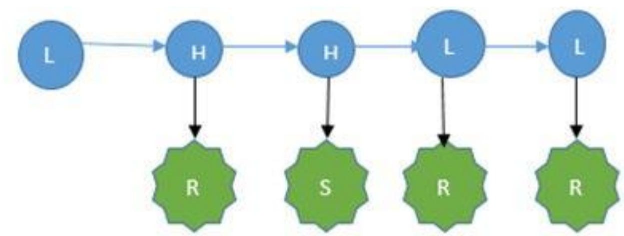
Find the probability of occurrence of this Pressure sequence observation: **S-S-R**

<u>Initialization Phase:</u>

$P(L)*P(S|L) = 0.5*0.2 = 0.1 \rightarrow 0.25$



$P(H)*P(S|H) = 0.5*0.6 = 0.3 \rightarrow 0.75$



Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

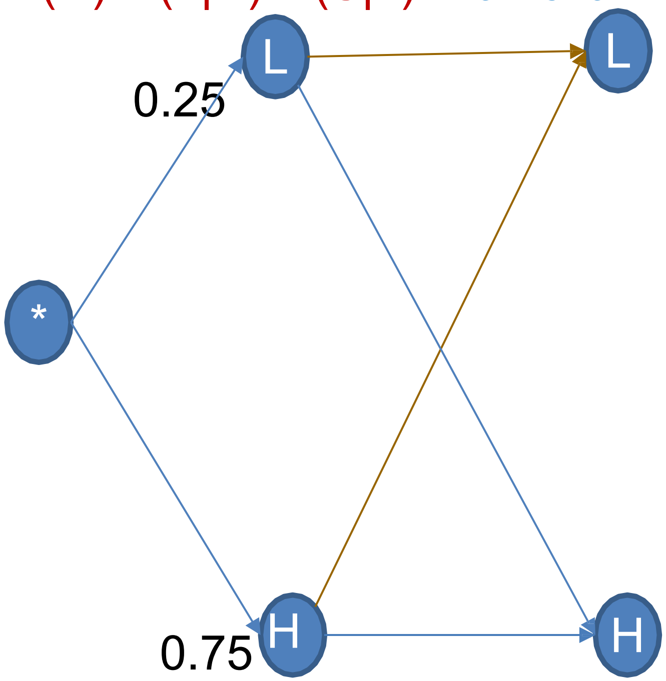
$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

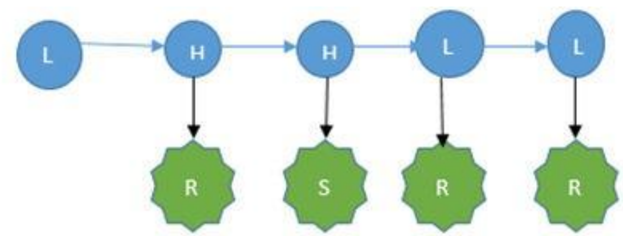
Hidden Markov Model

Forward Propagation Algorithm : S-S-R

P(L)*P(L|L)*P(S|L) = 0.25*0.5*0.2 = **0.025**

P(H)*P(L|H)*P(S|L) = 0.75*0.2*0.2 = **0.03**





Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

P(L)*P(H|L)*P(S|H) = 0.25*0.5*0.6 = **0.075**

P(H)*P(H|H)*P(S|H) = 0.75*0.8*0.6 = **0.36**

Evidence / Sensor Model/ Emission Probability Matrix

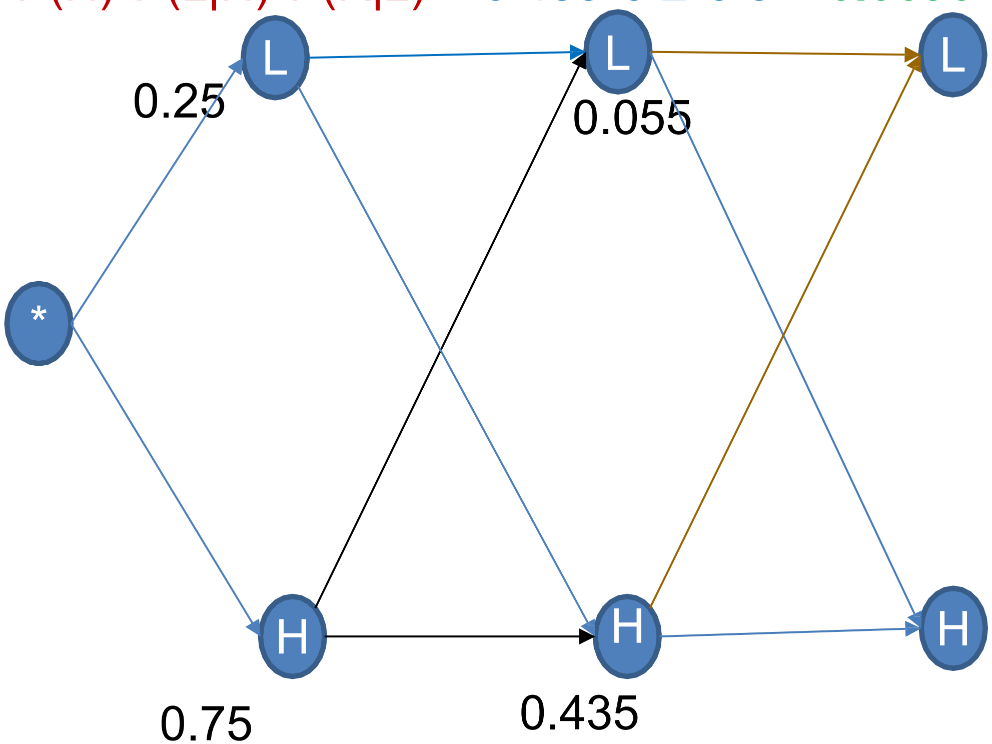
$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = \text{Rainy})$
0.2	0.6	$P(E_t = \text{Sunny})$

Hidden Markov Model

Forward Propagation Algorithm : S-S-R

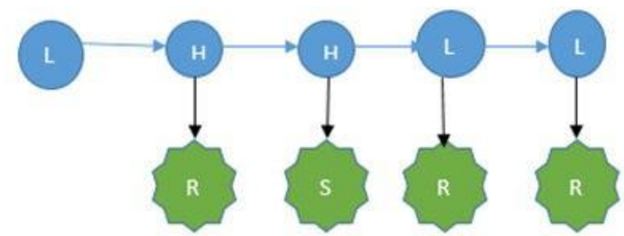
P(L)*P(L|L)*P(R|L) = 0.055*0.5*0.8 = **0.022**

P(H)*P(L|H)*P(R|L) = 0.435*0.2*0.8 = **0.0696**



P(L)*P(H|L)*P(R|H) = 0.055*0.5*0.4 = **0.011**

P(H)*P(H|H)*P(R|H) = 0.435*0.8*0.4 = **0.1392**



Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

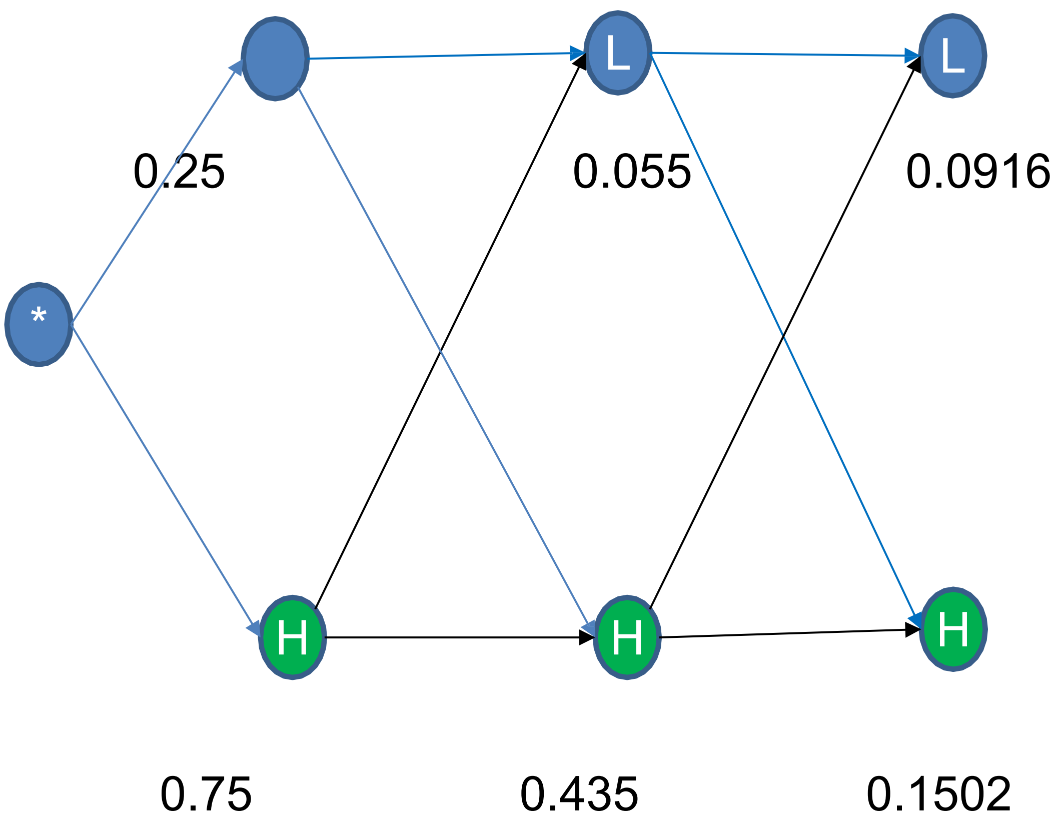
$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

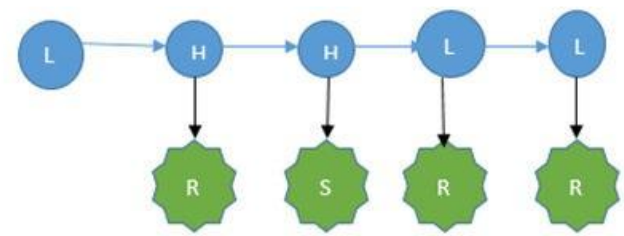
Hidden Markov Model

Forward Propagation Algorithm : S-S-R

Termination Phase:

P(S-S-R) = **0.2418**





Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

Hidden Morkov Model

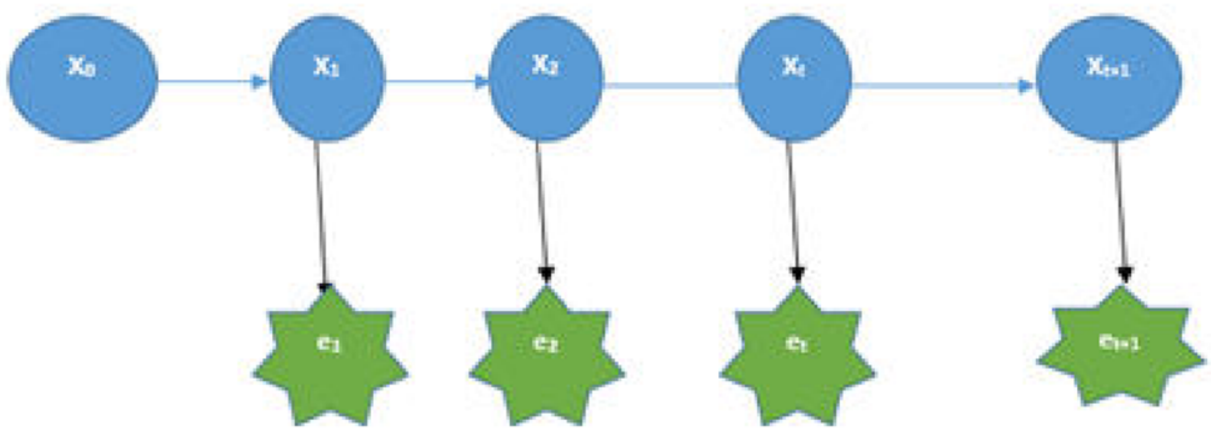
Inference: Type -3

Filtering : Forward Propagation Algorithm

Find the Current Pressure if sequence of weather observations recorded are: **S-S-R**

Intuition: $P(Xt_{+1} | E_{1...t+1}) = \boldsymbol{\alpha}\, P(et_{+1} | Xt_{+1}) * \sum_{X_t} P(X_{t+1}|X_t) * \boldsymbol{P}(\mathbf{X_t}| \mathbf{E_{1..t}})$

$$P(Xt_{+1} | E_{1...t+1}) = \boldsymbol{\alpha}\, P(e_{t+1} | Xt_{+1}) * \sum_{X_t} P(X_{t+1}|X_t) * \boldsymbol{P}(\mathbf{X_t}| \mathbf{E_{1..t}})$$



Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

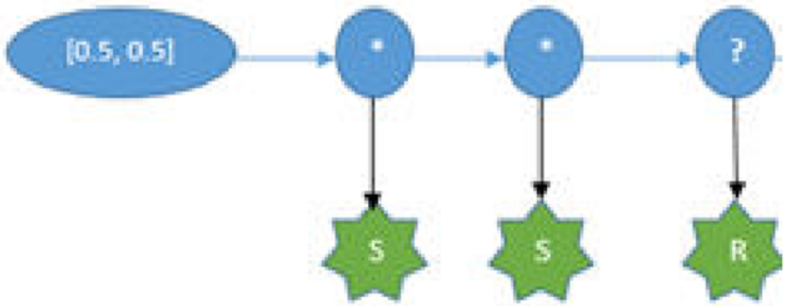
Hidden Morkov Model

Inference: Type -3

Filtering : Forward Propagation Algorithm

Find the Current Pressure if sequence of weather observations recorded are: **S-S-R**

Intuition: $P(E_{1...t}) = \sum_{i=1}^{N} P(E_{1...t} \mid X_{1...t}) * P(X_{1...t}) = \sum_{i=1}^{N} \prod_{j=1}^{t} P(Ej \mid Xj) * P(Xj \mid Xj_{-1})$



Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = \text{Rainy})$
0.2	0.6	$P(E_t = \text{Sunny})$

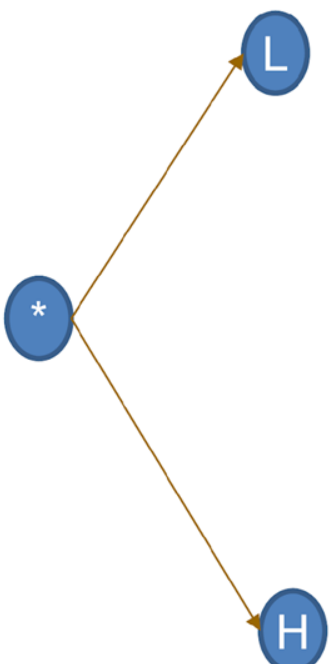
Hidden Morkov Model

Forward Propagation Algorithm

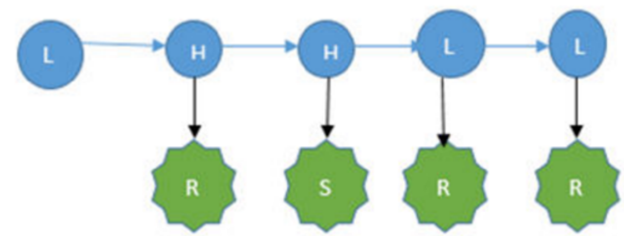
Pressure sequence observation: **S-S-R**

<u>Initialization Phase:</u>

P(L)*P(S|L) = 0.5*0.2 = 0.1 → 0.25



P(H)*P(S|H) = 0.5*0.6 = 0.3 → 0.75



Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

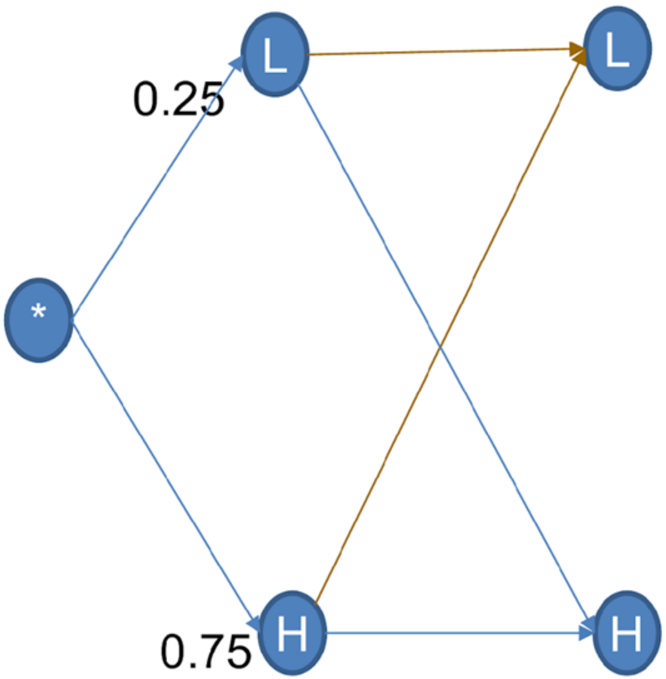
$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

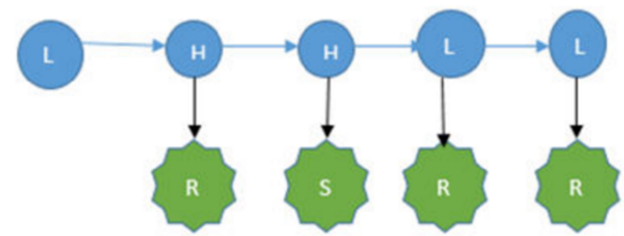
Hidden Morkov Model

Forward Propagation Algorithm : S-S-R

$P(L)*P(L|L)*P(S|L) = 0.25*0.5*0.2 = \mathbf{0.025}$

$P(H)*P(L|H)*P(S|L) = 0.75*0.2*0.2 = \mathbf{0.03}$





Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

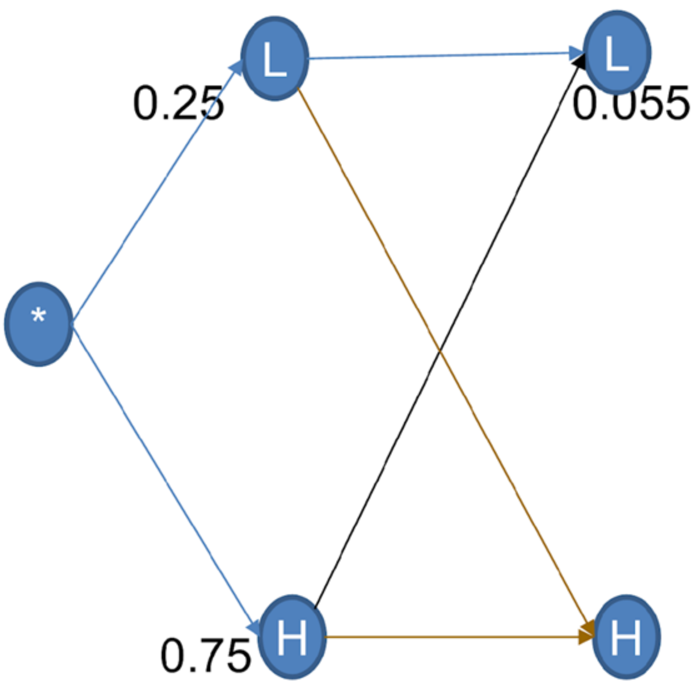
Evidence / Sensor Model/ Emission Probability Matrix

$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

Recursion Phase:

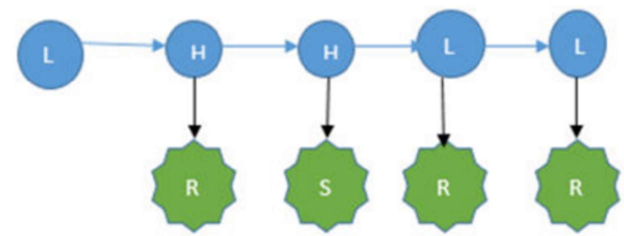
Hidden Morkov Model

Forward Propagation Algorithm : S-S-R



P(L)*P(H|L)*P(S|H) = 0.25*0.5*0.6 = **0.075**

P(H)*P(H|H)*P(S|H) = 0.75*0.8*0.6 = **0.36**



Transition Model / Probability Matrix

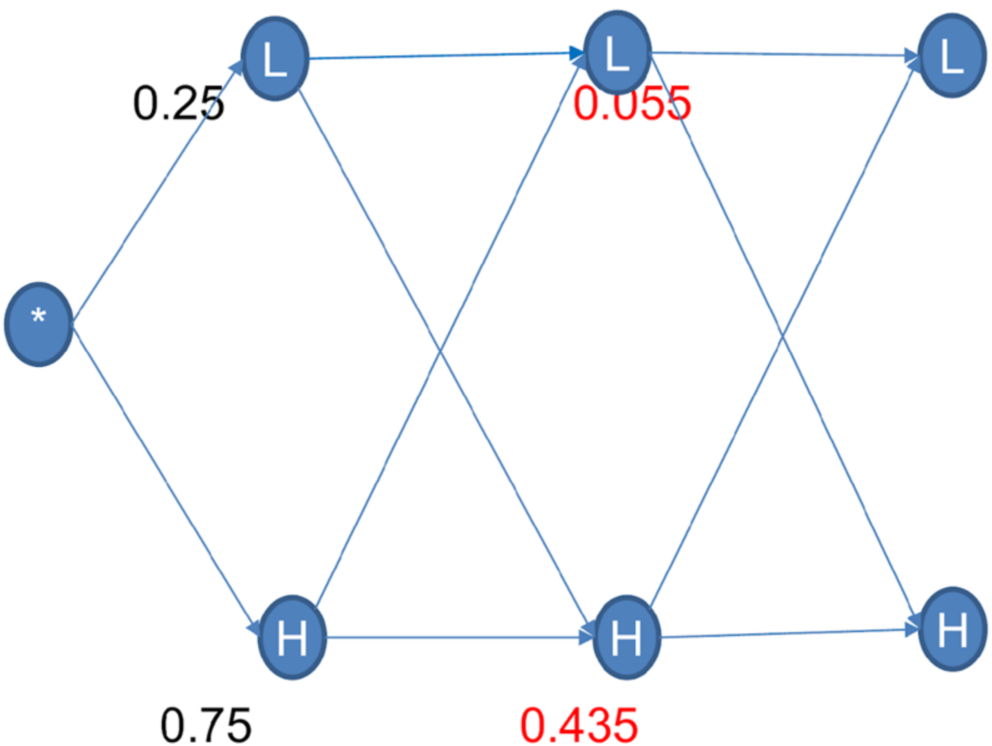
$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

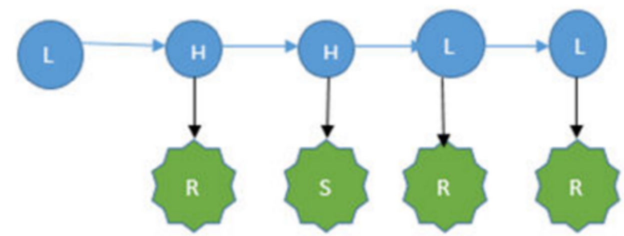
Evidence / Sensor Model/ Emission Probability Matrix

$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

Hidden Morkov Model

Forward Propagation Algorithm : S-S-R





Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

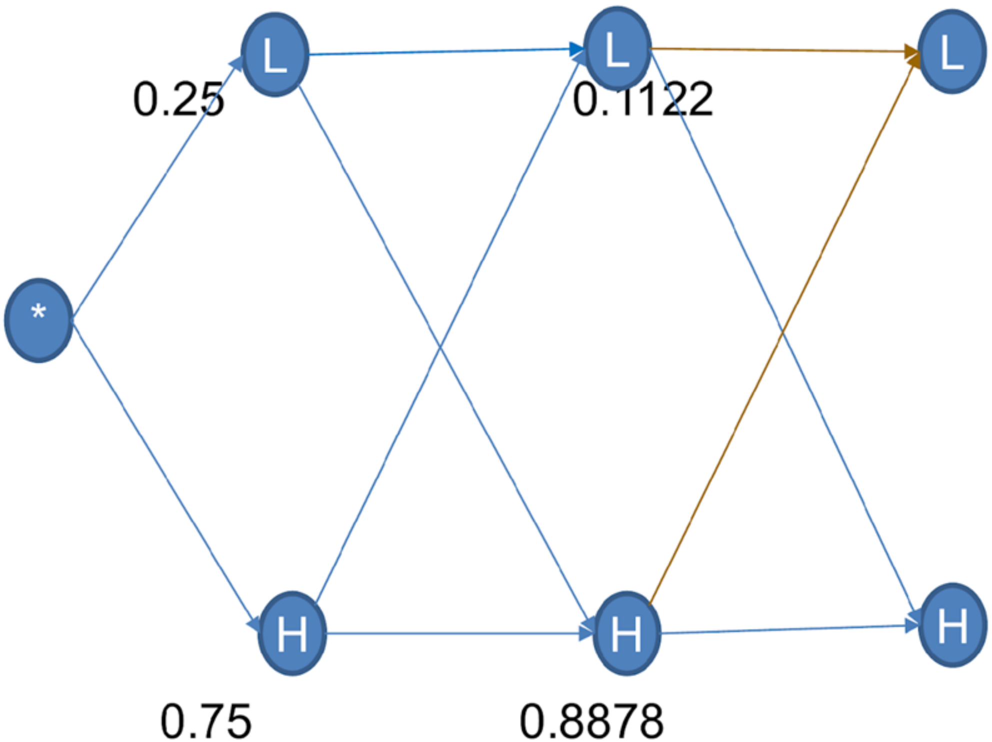
$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = \text{Rainy})$
0.2	0.6	$P(E_t = \text{Sunny})$

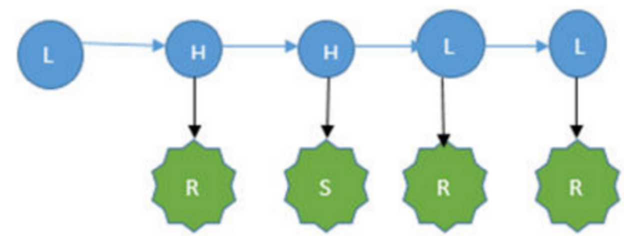
Hidden Morkov Model

Forward Propagation Algorithm : S-S-R

P(L)*P(L|L)*P(R|L) = 0.1122*0.5*0.8 = **0.04488**

P(H)*P(L|H)*P(R|L) = 0.8878*0.2*0.8 = **0.142048**





Transition Model / Probability Matrix

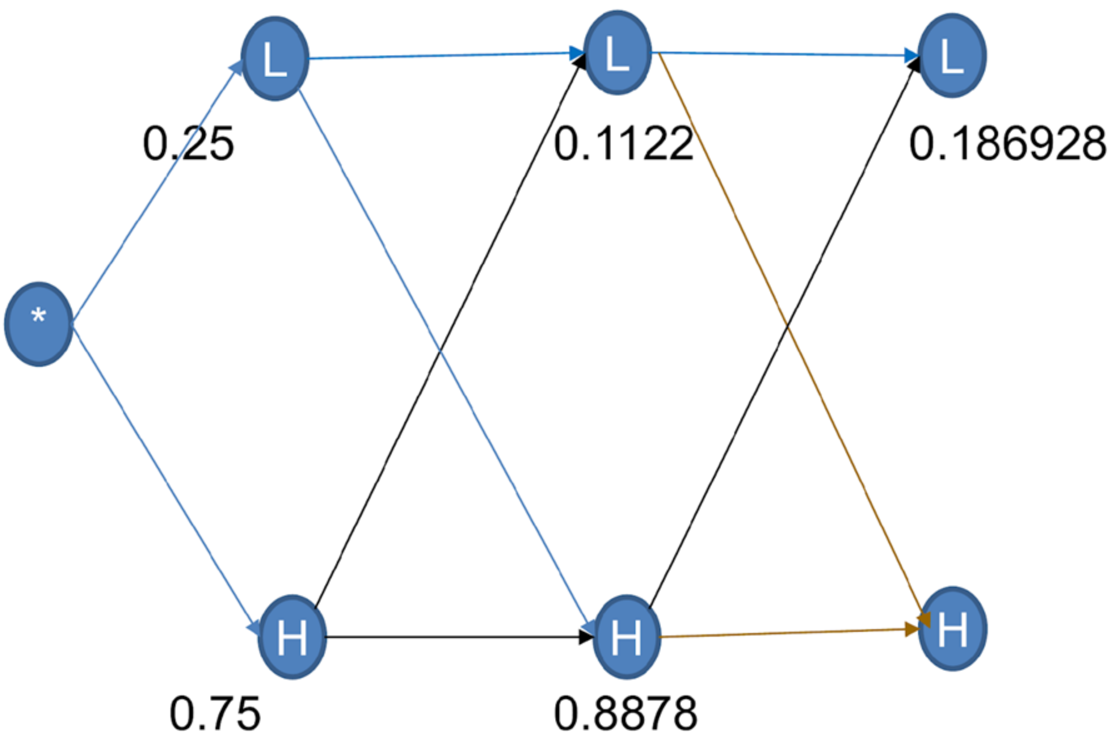
$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

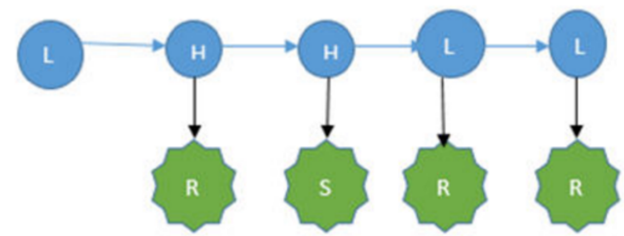
Hidden Morkov Model

Forward Propagation Algorithm : S-S-R



P(L)*P(H|L)*P(R|H) = 0.1122*0.5*0.4 = **0.02244**

P(H)*P(H|H)*P(R|H) = 0.8878*0.8*0.4 = **0.284096**



Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

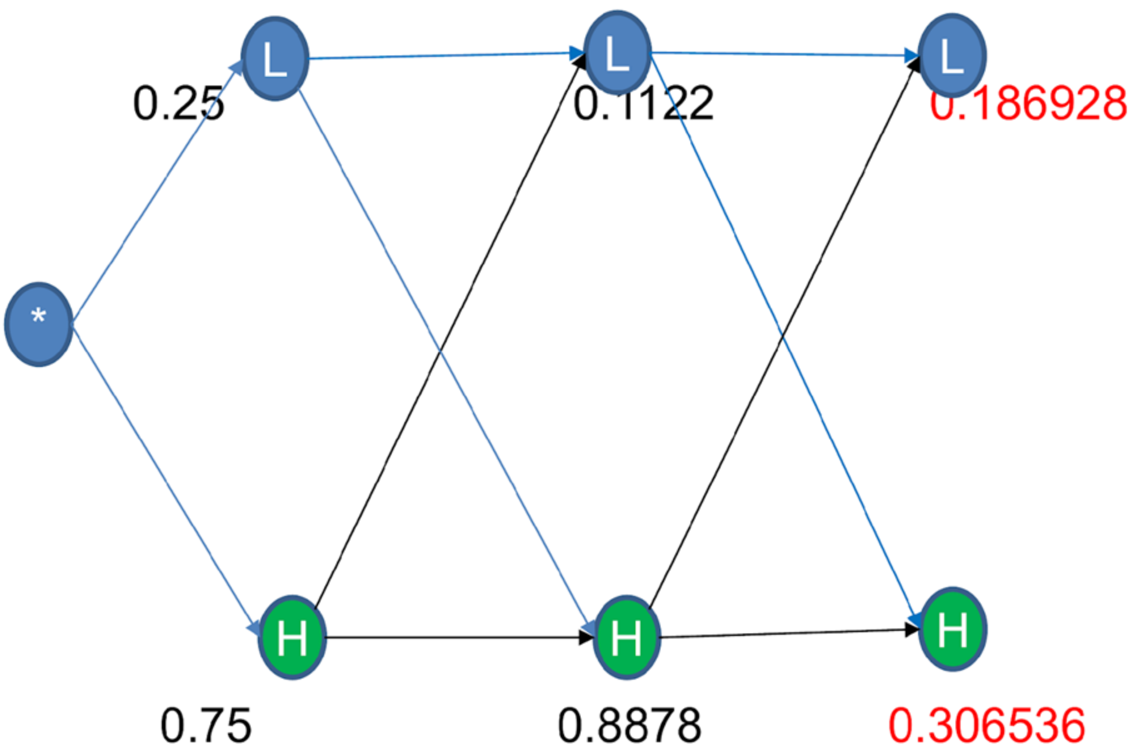
Evidence / Sensor Model/ Emission Probability Matrix

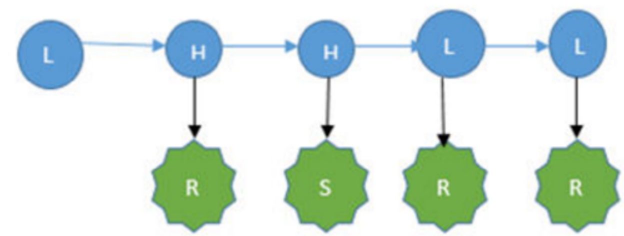
$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

Hidden Morkov Model

Forward Propagation Algorithm : S-S-R

Termination Phase:





Transition Model / Probability Matrix

P(U_{t-1} = HP)	P(U_{t-1} = LP)	←Previous
0.2	0.5	P(U_t = LP)
0.8	0.5	P(U_t = HP)

Evidence / Sensor Model/ Emission Probability Matrix

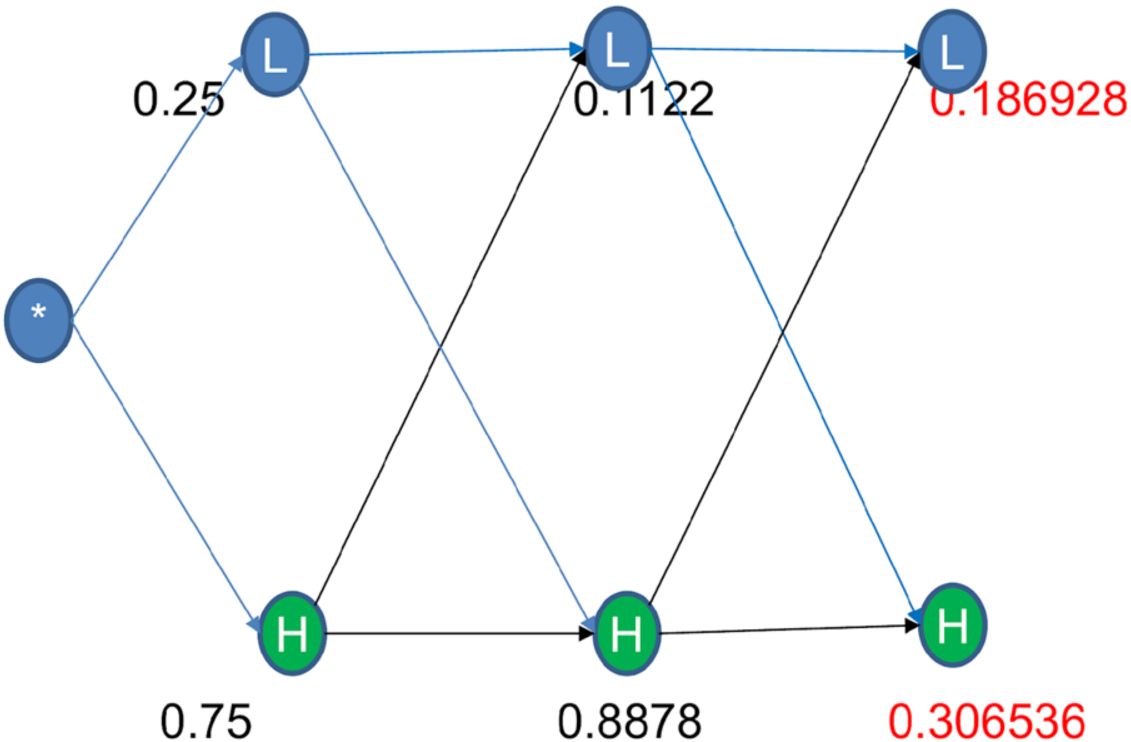
P(X_t = LP)	P(X_t = HP)	←Unobserved Evidence v
0.8	0.4	P(E_t = Rainy)
0.2	0.6	P(E_t = Sunny)

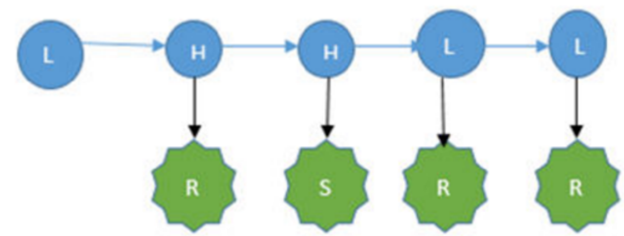
Hidden Morkov Model

Forward Propagation Algorithm : S-S-R

Termination Phase:

(0.37881,**0.62119**)





Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

Hidden Morkov Model

Inference: Type -2

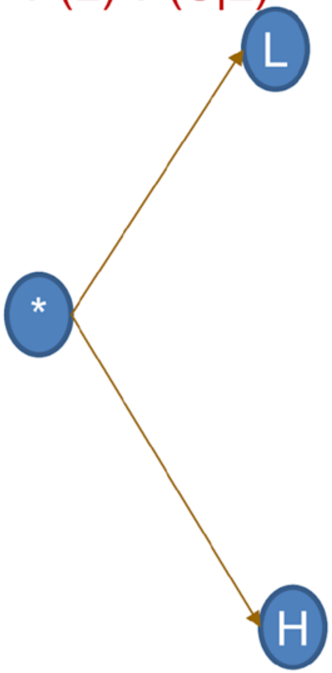
Most Likely Explanation : Veterbi Algorithm

Find the pattern in pressure that might have caused this observation: **S-S-R**

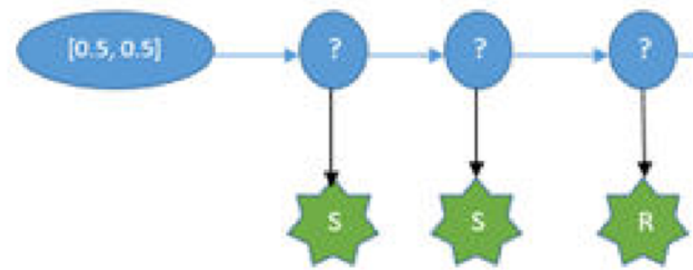
argmax $X_{1...t}$: $P(X_{1...t} \mid E_{1...t})$

MM Inf

P(L)*P(S|L) = 0.5*0.2 = 0.1 → 0.25



P(H)*P(S|H) = 0.5*0.6 = 0.3 → 0.75



Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

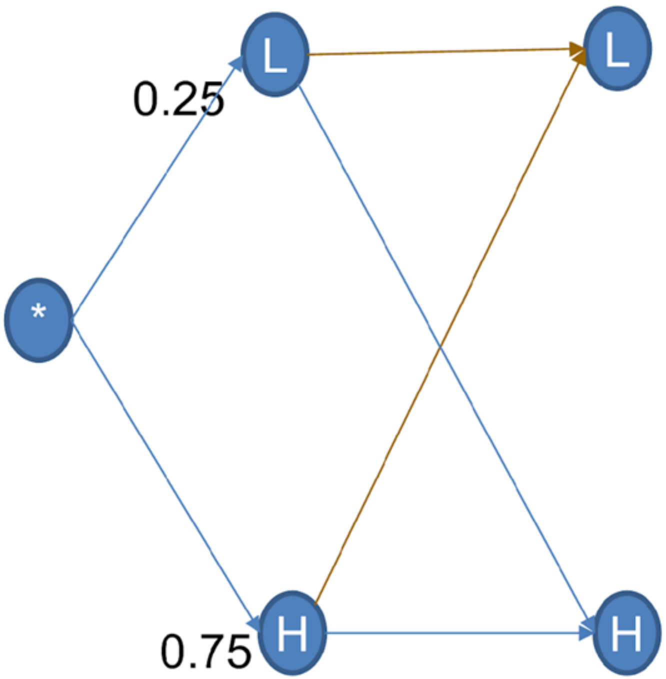
$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

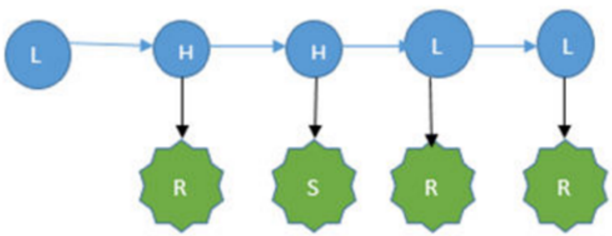
Hidden Morkov Model

Veterbi Algorithm : S-S-R

P(L)*P(L|L)*P(S|L) = 0.25*0.5*0.2 = 0.025

P(H)*P(L|H)*P(S|L) = 0.75*0.2*0.2 = **0.03**





Transition Model / Probability Matrix

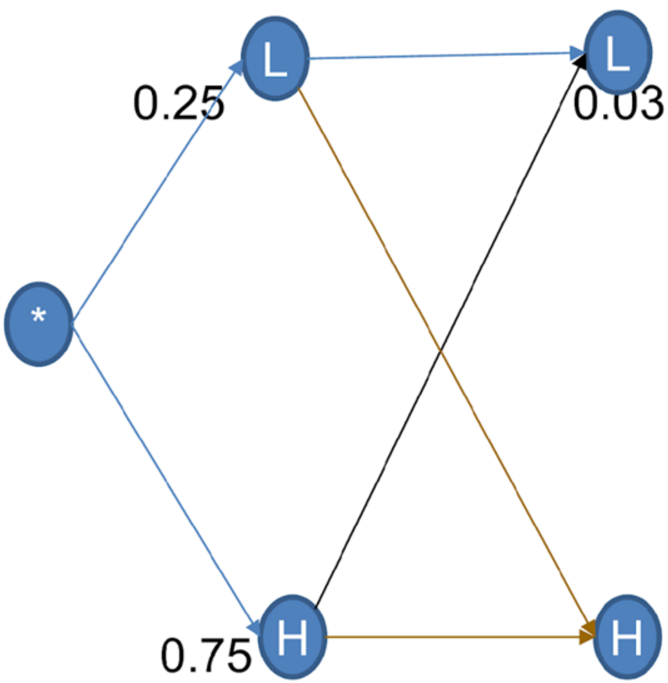
$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

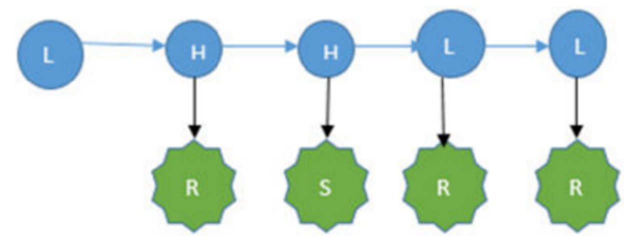
Evidence / Sensor Model/ Emission Probability Matrix

$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

Hidden Morkov Model

Veterbi Algorithm : S-S-R





P(L)*P(H|L)*P(S|H) = 0.25*0.5*0.6 = 0.075

P(H)*P(H|H)*P(S|H) = 0.75*0.8*0.6 = **0.36**

Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

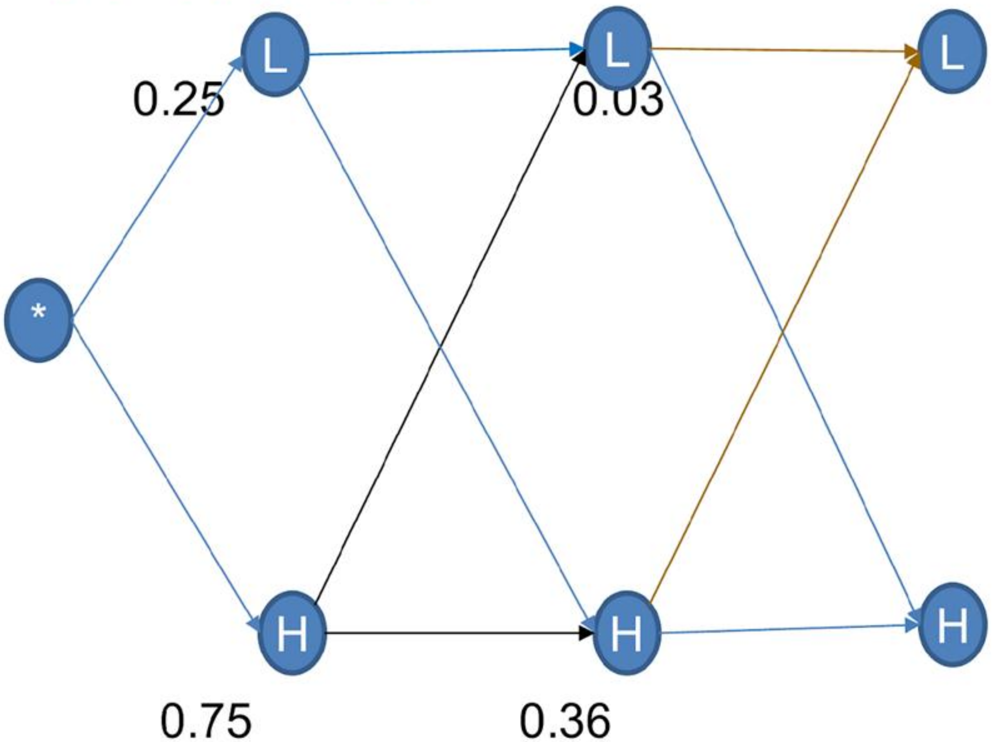
$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

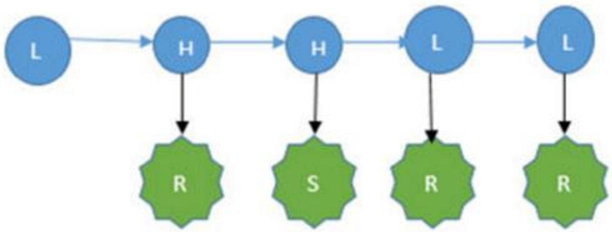
Hidden Morkov Model

Veterbi Algorithm : S-S-R

P(L)*P(L|L)*P(R|L) = 0.03*0.5*0.8 = 0.012

P(H)*P(L|H)*P(R|L) = 0.36*0.2*0.8 = **0.0576**





Transition Model / Probability Matrix

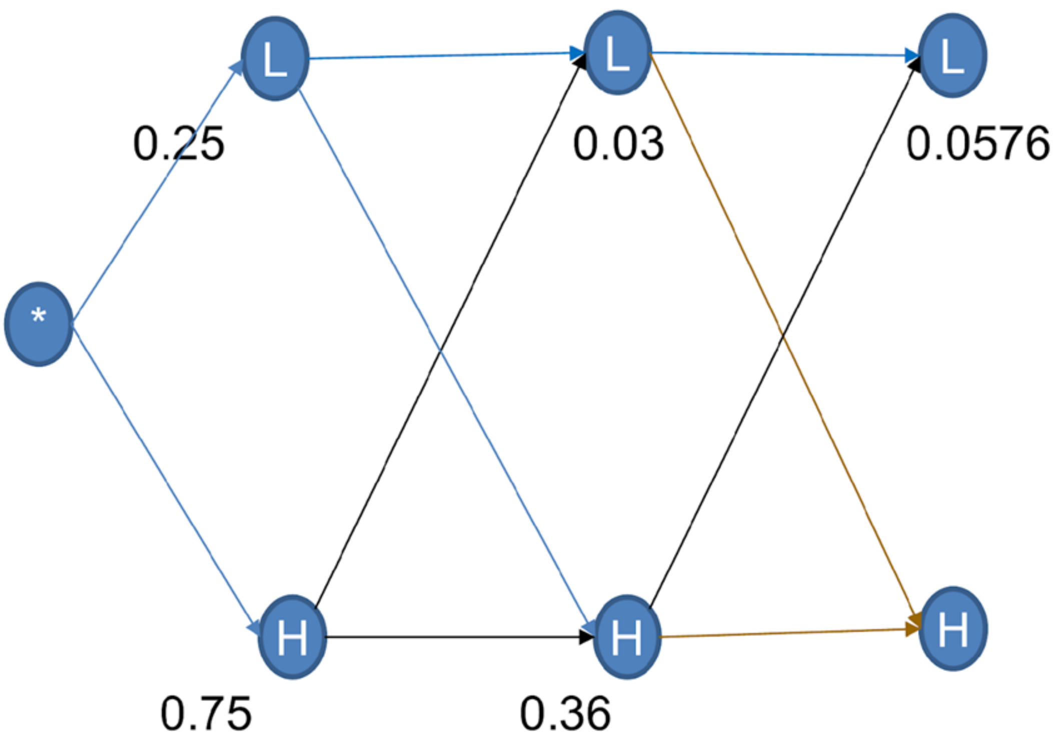
$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

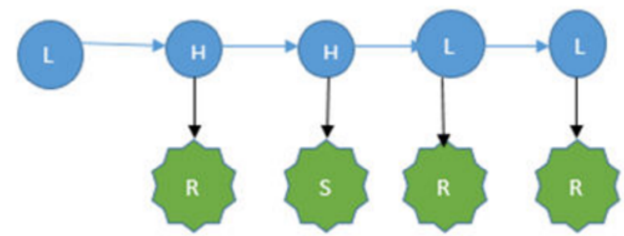
Hidden Morkov Model

Veterbi Algorithm : S-S-R



P(L)*P(H|L)*P(R|H) = 0.03*0.5*0.4 = 0.006

P(H)*P(H|H)*P(R|H) = 0.36*0.8*0.4 = **0.1152**



Transition Model / Probability Matrix

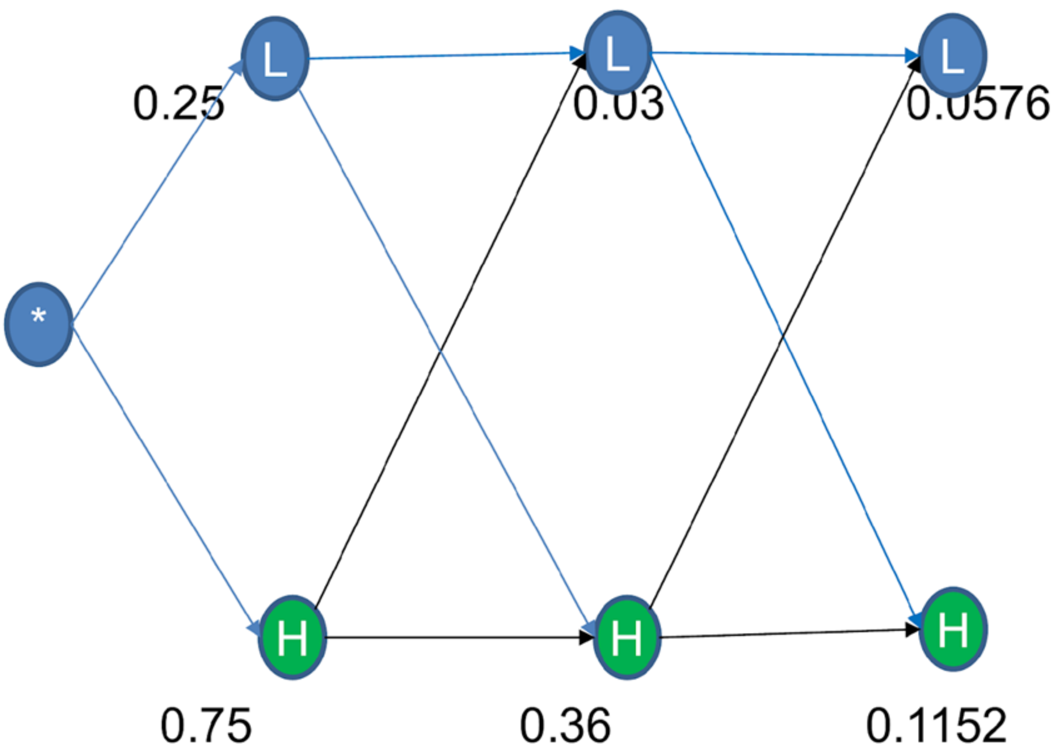
$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

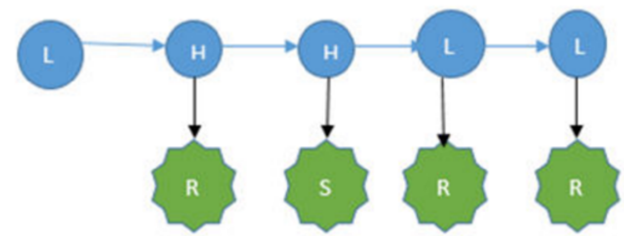
Evidence / Sensor Model/ Emission Probability Matrix

$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

Hidden Morkov Model

Veterbi Algorithm : S-S-R





Transition Model / Probability Matrix

$P(U_{t-1} = HP)$	$P(U_{t-1} = LP)$	←Previous
0.2	0.5	$P(U_t = LP)$
0.8	0.5	$P(U_t = HP)$

Evidence / Sensor Model/ Emission Probability Matrix

$P(X_t = LP)$	$P(X_t = HP)$	←Unobserved Evidence v
0.8	0.4	$P(E_t = Rainy)$
0.2	0.6	$P(E_t = Sunny)$

Explain HMM & Viterbi Algorithm

In general it's observed that low concentration/focus among students leads to low grades in 80% of time and high concentration/focus produces good grades in 95% of time. 60% of time students realize that low concentration had led to challenges in completed exams and they decide to concentrate more in the upcoming next exams. 30% of time students are always prudent and proactively prepare with concentration for every exams. At the start of the course enrolment, usually 98% of students are willing to concentrate more for evaluations.

AI agent is designed to perform learner behavioral analysis based on given model. With inputs of grade scored by students for few semesters, how the AI system answers the query, where B=Scoring Low Grade, G= Scoring High Grade.

Example Markov Model Query :
"What is the most likely explanation of the student behavior if pattern observed is B-G-B"? – Apply Veterbi Algorithm

Citations and References

- Few content for these slides may have been obtained from prescribed books and various other source on the Internet
- I hereby acknowledge all the contributors for their material and inputs and gratefully acknowledge people others who made their course materials freely available online.
- .I have provided source information wherever necessary
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- I have added and modified the content to suit the requirements of the class dynamics & live session's lecture delivery flow for presentation

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Thank You for your time & attention !

Contact : parthasarathypd@wilp.bits-pilani.ac.in